

PGR Platform

Postgraduate Research Student Lifecycle Management

User Manual — What each screen does, why it exists, and how to use it.

v2 · 2026-08-24 · built against the running app with real screenshots

Backend	FastAPI · SQLAlchemy 2.0 async · PostgreSQL 18
Frontend	Next.js 14 · Tailwind · shadcn/ui Redwood Professional
Modules	22 domain modules · 168 API endpoints · 26 UI routes
Tests	224 passing (unit, integration, end-to-end)
Migration head	800191647194
Demo login	admin@example.com / admin123 (Institution Admin)

How to read this manual

Each section covers one screen or workflow in the platform. Every section answers three questions:

What it solves — the real-world problem this screen exists to remove.

How to use it — the exact clicks or fields, in order.

Screenshot — captured live from the running app so you can match it to what you see.

Getting in

Launch the platform with `start-all.bat`. That single script starts PostgreSQL, the backend API on port 8000, the background worker (which fires scheduled jobs, dispatches the outbox, and delivers notifications), the frontend on port 3000, and opens your browser at the login page.

The demo institution ships with two accounts you can sign in with straight away:

Role	Email	Password	Sees
Institution Administrator	admin@example.com	admin123	Everything, including settings
Supervisor	elena.ford@example.com	super123	Only their own supervisees

Sign in

What it solves. One entry point for every persona (administrators, supervisors, students, executives). The credentials passed here are exchanged for a 15-minute access JWT plus a 14-day refresh token; the app renews the access token silently, so you almost never see this screen twice in a day.

How to use it.

- Type your work email.
- Type your password. Five failed attempts locks the account for 15 minutes — use *Forgot password* to recover.
- Click **Sign in**. You will land on the Executive dashboard by default; students land on *My journey*.



PGR Platform

Sign in to continue

Email

Password

[Forgot password?](#)

Sign in

Demo: admin@example.com / admin123

Login page. Notice the pre-filled demo credentials — clear them before signing in as a real user.

1 · The lifecycle at a glance

The PGR platform models one continuous **student journey**: a person appears as a *research demand* or an *applicant*, moves into an *application*, is issued an *offer*, accepts, becomes a *student*, is examined, and — one atomic transaction — graduates into *alumnus*. The same `person_id` is carried through every step so you never lose the thread.

Executive dashboard

What it solves. The one screen leadership opens first thing in the morning. Six lifecycle metrics (active researchers, applications in pipeline, applicant→student conversion, funded students, theses under exam, completions) and four administrator queues (assessments awaiting decision, offers to accept, progression reviews due, theses to examine) sit side-by-side so nothing that needs attention today can hide.

How to use it.

- Read the six **Lifecycle metrics** tiles for an at-a-glance state of the school.
- Scan the **Administrator queues** — non-zero counts are work waiting on a human decision.
- Use the sidebar to drill into any domain area.
- Click **Ask PGR** (bottom-right, or press **Ctrl/Cmd+K**) for natural-language queries.

The screenshot displays the PGR Platform Executive dashboard. The interface is divided into a left sidebar, a top header, and a main content area.

Header: PGR Platform RESEARCH LIFECYCLE, Fusion Practices Oracle Partner, PGR Platform, Search or jump to..., Ctrl+K, and a notification bell.

Sidebar (Workspace): Dashboard (selected), Analytics, My journey, Tasks, Persons, Research, Recruitment, Admissions, Students, Supervision, Progression, Funding, Thesis, Completion.

Sidebar (Administration): Funding integrity, Statutory, Workflows.

Main Content Area:

- Platform connectivity:** API reachable, Database ok, Signed in as admin@example.com, Role Institution Administrator, PGR Administrator.
- Lifecycle metrics:**
 - ACTIVE RESEARCHERS: 266 (registered / active)
 - APPLICATIONS: 344 (in pipeline)
 - CONVERSION RATE: 99.7% (applicant → student)
 - FUNDED STUDENTS: 2 (active arrangement)
 - THESES SUBMITTED: 68 (awaiting/under exam)
 - COMPLETIONS: 31 (graduated)
- Administrator queues:**
 - AWAITING ASSESSMENT: 1 (applications)
 - OFFERS TO ACCEPT: 0 (issued)
 - REVIEWS DUE: 0 (progression)
 - THESES SUBMITTED: 68 (to examine)

Bottom Right: Ask PGR button with a keyboard shortcut icon.

Executive dashboard — the platform's front page.

2 · Before a student exists

Most SIS platforms start at the application. The PGR platform starts one step earlier — **research awards** (grant references, mastered outside the platform) and **research demand** (posts the school has committed to open). Every subsequent position, applicant and student can be traced back to the award and the demand that created it.

Research — awards and demand

What it solves. Answers *where did this PhD post come from and how much money is behind it?* Awards are read-only when mastered by Research Office (the little padlock); demand rows move through a small state machine (proposed → approved → posted). Later, when a position is opened, this is where the provenance chain starts.

How to use it.

- **Awards** tab: browse award references pulled in via signed webhook from Research Office.
- **Demand** tab: create a new demand, attach it to an award (or leave standalone), then approve → post.
- Follow a demand row into Recruitment to see the opportunity it produced.

PGR Platform RESEARCH LIFECYCLE

FUSION PRACTICES ORACLE Partner **PGR Platform**

Q Search or jump to... CtrlK

WORKSPACE

- Dashboard
- Analytics
- My journey
- Tasks
- Persons
- Research**
- Recruitment
- Admissions
- Students
- Supervision
- Progression
- Funding
- Thesis
- Completion

ADMINISTRATION

- Funding integrity
- Statutory
- Workflows

Research

Where a PGR position comes from: an award funds a demand, a demand becomes a position.

Research demand Awards Supervisor match Relationship map

Demand is the recorded *need* for a researcher. It becomes a recruitable position only once it is approved and positioned. **+ Raise demand**

Title	Award	Places	Target start	Status
PGR researcher for Robotics Programme Grant	SCALE-AWD-99-020	2	2026-11-20	positioned Move to...
PGR researcher for Climate Systems Programme Grant	SCALE-AWD-99-019	3	2026-11-20	positioned Move to...
PGR researcher for Machine Learning Programme Grant	SCALE-AWD-99-018	3	2026-11-20	positioned Move to...
PGR researcher for Health Informatics Programme Grant	SCALE-AWD-99-017	2	2026-11-20	positioned Move to...
PGR researcher for Robotics Programme Grant	SCALE-AWD-99-016	2	2026-11-20	positioned Move to...
PGR researcher for Computational Biology Programme Grant	SCALE-AWD-99-015	2	2026-11-20	positioned Move to...
PGR researcher for Machine Learning Programme Grant	SCALE-AWD-99-014	3	2026-11-20	positioned Move to...
PGR researcher for Materials Science Programme Grant	SCALE-AWD-99-013	1	2026-11-20	positioned Move to...
PGR researcher for Robotics Programme Grant	SCALE-AWD-99-012	2	2026-11-20	positioned Move to...
PGR researcher for Materials Science Programme Grant	SCALE-AWD-99-011	3	2026-11-20	positioned Move to...
PGR researcher for Materials Science Programme Grant	SCALE-AWD-99-010	3	2026-11-20	positioned Move to...
PGR researcher for Materials Science Programme Grant	SCALE-AWD-99-009	2	2026-11-20	positioned Move to...
PGR researcher for Climate Systems Programme Grant	SCALE-AWD-99-008	2	2026-11-20	positioned Move to...

admin admin@example.com

Ask PGR

Research: awards on the left, demand on the right, both feeding recruitment downstream.

Recruitment — opportunities and applications

What it solves. Every posted position, with **places filled / available** shown live and a **Provenance** dialog that walks award → demand → position → shortlist. The pipeline supports the two entry routes the CIO asked for: **Route A** (apply to an advertised opportunity) and **Route B** (applicant proposes their own topic; opportunity FK is null, and a proposal document reference records their intent).

How to use it.

- Filter opportunities by department, area or status.
- Click an opportunity to open its detail; the **Provenance** button shows what created it.
- Under **Applications**, click any row to open the assessment history and route badge.

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RESEARCH LIFECYCLE

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PGR Platform

Q Search or jump to... CtrlK

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WORKSPACE

Dashboard

Analytics

My journey

Tasks

Persons

Research

Recruitment

Admissions

Students

Supervision

Progression

Funding

Thesis

Completion

ADMINISTRATION

Funding integrity

Statutory

Workflows

admin
admin@example.com

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➡️

🏠 > Recruitment

Recruitment

Opportunities and the application pipeline.

Opportunities

Applications

+ New opportunity

Title	Stipend	Places	Status	Provenance
PhD in Computational Biology	GBP 18,500	16 / 2 full	open Move to...	🔗 Lineage
PhD in Computational Biology	GBP 19,000	12 / 3 full	open Move to...	🔗 Lineage
PhD in Health Informatics	GBP 19,000	12 / 3 full	open Move to...	🔗 Lineage
PhD in Health Informatics	GBP 19,500	8 / 2 full	open Move to...	🔗 Lineage
PhD in Computational Biology	GBP 20,500	13 / 2 full	open Move to...	🔗 Lineage
PhD in Computational Biology	GBP 18,000	8 / 2 full	open Move to...	🔗 Lineage
PhD in Robotics	GBP 20,000	12 / 3 full	open Move to...	🔗 Lineage
PhD in Materials Science	GBP 19,500	12 / 1 full	open Move to...	🔗 Lineage
PhD in Machine Learning	GBP 20,000	11 / 2 full	open Move to...	🔗 Lineage
PhD in Computational Biology	GBP 20,000	12 / 3 full	open Move to...	🔗 Lineage
PhD in Computational Biology	GBP 18,500	10 / 3 full	open Move to...	🔗 Lineage
PhD in Computational Biology	GBP 19,000	18 / 2 full	open Move to...	🔗 Lineage

🔗 Ask PGR

🔍

Recruitment pipeline. Places available/filled prevents over-recruitment at the row level.

Application detail

What it solves. Everything about a single applicant: stage history, assessments, referees, and — critically — the **route** they came in on. When the pipeline chair is ready, **Recommend for offer** hands the application to Admissions in a single click without leaving this page.

How to use it.

- Open an application row.
- Review the stage-history timeline; log a new assessment via the button in the assessments card.
- When ready, click **Recommend for offer** — this creates the offer draft in Admissions.

The screenshot displays the 'Application detail' page in the PGR Platform. The left sidebar contains navigation links for Workspace (Dashboard, Analytics, My journey, Tasks, Persons, Research, Recruitment, Admissions, Students, Supervision, Progression, Funding, Thesis, Completion) and Administration (Funding integrity, Statutory, Workflows). The main content area shows the application details for 'Omar Okafor', with a route of 'opportunity led' and a stage of 'converted'. It includes sections for 'Advance stage' (with a target stage dropdown and a reason field), 'Record assessment' (with a decision dropdown set to 'recommended' and a rationale field), 'Offer' (with a 'Create offer' button), and 'Stage history' (showing no history). A 'Back to recruitment' link is also present. The bottom right corner features an 'Ask PGR' button.

Application detail — stage timeline, assessments, and the recommend-for-offer action.

Admissions — offers

What it solves. The gap between an internal decision and a signed student. Offers move through **draft** → **issued** → **accepted/declined** → **withdrawn**. **Accepting an offer creates the student atomically** — the expected end date is derived, the research project is linked to the award (so funding lineage will hold), and the person's *applicant* identity closes as the *student* identity opens.

How to use it.

- Find the offer in **Issued**.
- Fill any missing conditions and click **Issue** if it is still a draft.
- Once the applicant replies, click **Accept** or **Decline**. Accepting creates the student record.

3 · The active student record

Once accepted, every student has a single record that carries the whole timeline: supervision, meetings, milestones, funding, thesis, corrections, documents and audit history. Nothing lives in a spreadsheet.

Students — list and search

What it solves. Every registered PGR in one filterable list. Row scoping means a supervisor sees only their own supervisees on this screen; an administrator sees all. Out-of-scope reads return *not found*, so a supervisor cannot even discover that a student in another school exists.

How to use it.

- Type into the search bar to filter by name or student reference.
- Use the department and status filters at the top of the table.
- Click any row to open the full student record.

The screenshot displays the 'Students' list in the PGR Platform. The interface includes a sidebar with navigation links, a top header with a search bar, and a main content area with a table of student records. The table has four columns: 'Student ref', 'Status', 'Mode', and 'Start'. The 'Status' column uses color-coded labels: green for 'active', orange for 'withdrawn', yellow for 'registered', and red for 'suspended'. The 'Mode' column lists 'full time' or 'part time'. The 'Start' column shows dates. A bottom right button labeled 'Ask PGR' is visible.

Student ref	Status	Mode	Start
SCALE-99-0300	active	full time	2025-09-04
SCALE-99-0299	withdrawn	full time	2022-09-21
SCALE-99-0298	withdrawn	full time	2026-05-24
SCALE-99-0297	withdrawn	part time	2023-03-29
SCALE-99-0296	active	full time	2026-05-29
SCALE-99-0295	active	full time	2025-10-16
SCALE-99-0294	active	full time	2024-06-11
SCALE-99-0293	active	full time	2023-02-25
SCALE-99-0292	withdrawn	full time	2025-02-12
SCALE-99-0291	registered	full time	2023-04-27
SCALE-99-0290	active	part time	2025-12-31
SCALE-99-0289	active	full time	2024-10-22
SCALE-99-0288	active	full time	2025-09-21
SCALE-99-0287	registered	full time	2023-01-06
SCALE-99-0286	suspended	full time	2022-10-12
SCALE-99-0285	active	full time	2026-07-14
SCALE-99-0284	active	full time	2025-10-02

Students list. Every row leads to the record on the next page.

Student record — the marquee screen

What it solves. The single richest screen in the platform. Down the page: **Record** (identifiers and mode), **Lifecycle changes** (suspensions and extensions with the original expected-end preserved), **Pattern Lab prediction** (governed ML advisory that sits *beside* the deterministic risk flags, never replaces them), **Supervisors, Meetings, Milestones + panels + appeals, Funding + payments + waivers** with a **Lineage** dialog, **Thesis** with viva and corrections, **Documents**, and **History** (the audit trail).

How to use it.

- Every section is collapsible — click the header to fold sections you don't need.

- **Request...** in Lifecycle changes opens the suspension/extension flow. Approval by a second admin is required before the timeline recalculates.
- **Lineage** beside the funding table shows award → arrangement → payments, and names any gap.

The screenshot displays the PGR Platform interface. On the left is a sidebar with navigation options under 'WORKSPACE' (Dashboard, Analytics, My journey, Tasks, Persons, Research, Recruitment, Admissions, Students, Supervision, Progression, Funding, Thesis, Completion) and 'ADMINISTRATION' (Funding integrity, Statutory, Workflows). The main content area shows the 'Students' page for 'Omar Okafor'. It includes a 'Record' section with student details (STUDENT REF: SCALE-99-0300, STATUS: active, STUDY MODE: full time, START DATE: 2025-09-04, EXPECTED END: 2029-03-05, RESEARCH TOPIC: Health Informatics: modelling study). Below this is a 'Lifecycle changes' section with a '+ Request...' button and a note about suspensions. The 'Predicted risk (Pattern Lab)' section shows a 'Funding Continuity Risk model' with a 6% risk level and a bar chart. A bottom bar shows the user 'admin' and a search bar.

Student record — everything about one student, one page. Suspended students are excluded from chase-ups.

Persons

What it solves. The identity spine of the platform. One person_id ties every stage together and, uniquely, **concurrent identities** are supported — a PGR who is also a staff member has both identities open at the same time.

How to use it.

- Search by name, email, or external ID.
- Click a person row for the timeline of identities (applicant → student → alumnus, plus employee if any).
- Open/close identity actions live on the person detail — closing a student identity *without* opening a new one is what graduation does automatically.

The screenshot shows the PGR Platform interface. On the left is a sidebar with a 'WORKSPACE' section containing links to Dashboard, Analytics, My journey, Tasks, and Persons (which is highlighted). Below this is an 'ADMINISTRATION' section with links to Funding integrity, Statutory, and Workflows. The main content area is titled 'Persons' and includes a search bar 'Search by name or email...'. Below the search bar is a table with the following columns: Name, Email, Nationality, and Current identities. The table lists 18 individuals, including Amara Abara, Chen Abara, Dara Abara, Iris Abara, Jonas Abara, Priya Abara, Quinn Abara, Sami Abara, Tara Abara, Uma Abara, Viktor Abara, Marcus Bell, Ben Bennett, and Chen Bennett. Each row shows their email and nationality, and a 'Current identities' column with green buttons labeled 'student · current'. Marcus Bell and Ben Bennett have additional orange buttons labeled 'employee · current'. At the bottom right of the table is a blue button labeled 'Ask PGR' with a magnifying glass icon.

Name	Email	Nationality	Current identities
Amara Abara	scale99s144@student.example.ac.uk	Indian	student · current
Chen Abara	scale99s271@student.example.ac.uk	Brazilian	student · current
Dara Abara	scale99s214@student.example.ac.uk	Chinese	student · current
Iris Abara	scale99s109@student.example.ac.uk	Kenyan	student · current
Iris Abara	scale99s33@student.example.ac.uk	Brazilian	student · current
Jonas Abara	scale99s150@student.example.ac.uk	German	student · current
Priya Abara	scale99s218@student.example.ac.uk	German	student · current
Priya Abara	scale99s200@student.example.ac.uk	—	student · current
Quinn Abara	scale99s23@student.example.ac.uk	Indian	student · current
Sami Abara	scale99s259@student.example.ac.uk	British	student · current
Tara Abara	scale99s143@student.example.ac.uk	Kenyan	student · current
Uma Abara	scale99s184@student.example.ac.uk	Irish	student · current
Viktor Abara	scale99s195@student.example.ac.uk	Chinese	student · current
Marcus Bell	marcus.bell@example.ac.uk	Irish	student · current employee · current
Ben Bennett	sup9934@uni.example.ac.uk	British	employee · current
Chen Bennett	scale99s84@student.example.ac.uk	British	student · current

Persons list — a person can carry any combination of applicant/student/employee/alumnus identities.

Person detail — 360 timeline

What it solves. Everything the platform knows about one human, chronological. Used both by administrators looking up a person and by the alumni team confirming continuity from student to alumnus.

How to use it.

- Scroll the timeline to see every identity opening and closing, with source system and effective dates.
- Add a relationship (parent/child, spouse) if it matters for right-to-contact rules.

[← Back to persons](#)

1985-11-09

One person_id, several identities held at once — applicant, student, employee, alumni, researcher.

Close

Identities are additive. Closing one dates it and leaves it in place: the others carry on, so a person's history is never rewritten to fit their current role.

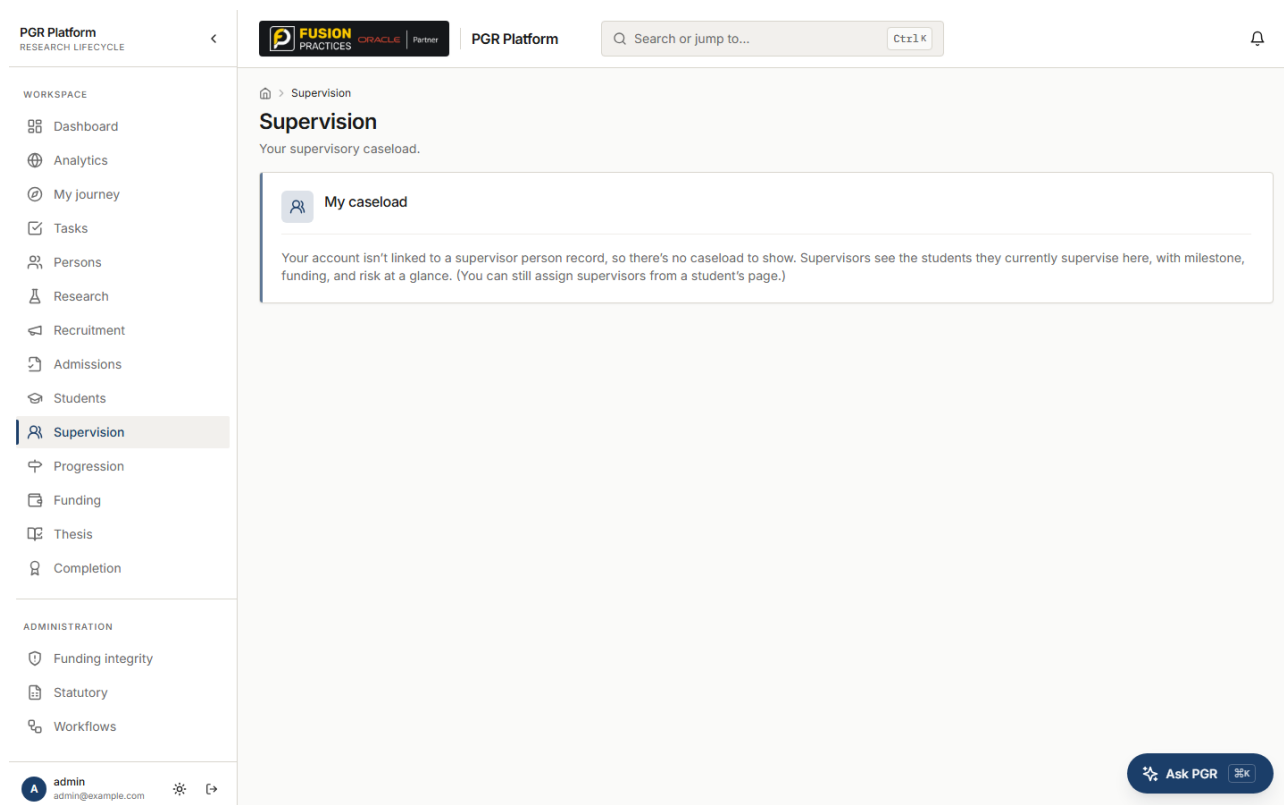
- Student (2025-05-03 to current)
2025-05-03

Supervision

What it solves. The caseload view. Each supervisor sees their supervisees with **last meeting** and an **overdue** column driven by the institutional expected-meeting interval (default 90 days, editable in Settings). The capacity guard prevents assigning a supervisor beyond the institutional maximum (default 8).

How to use it.

- Filter by department or search by student.
- Overdue rows are highlighted — click into the student and use **Log meeting**.
- Assign or end a supervision relationship from the student record; history is preserved, never overwritten.



Supervision caseload. Suspended students do not raise overdue flags.

Progression

What it solves. Configurable milestones per programme (upgrade, annual review, etc.), **review panels** (chair plus one independent assessor who is not a supervisor), mandatory conditions with auto re-review, outcome letters, and **appeals** that reopen a decided milestone.

How to use it.

- Milestones are generated for you on the scheduler (default every 60 seconds).
- Convene the panel from the student record; the platform enforces the independent-assessor rule.
- Record the outcome — pass / conditions / fail — and the appeal window opens automatically for 28 days.

Funding

What it solves. Funding over time, not a single field on the student record. A student can hold several arrangements in sequence or in parallel; each has a **stipend payment schedule** that flows: *due* → *approve* → *paid*. Approving a payment emits a domain event to Finance via the outbox. Fee waivers hang off the funding record and are logged and audited.

How to use it.

- Filter by source or by student.
- Approve a due stipend inline; the outbox event is written in the same transaction as the row change.
- Add a fee waiver by opening the student's funding card and clicking **Waive fee**.

The screenshot shows the PGR Platform interface. The top navigation bar includes the PGR Platform logo, a search bar, and a notification bell. The left sidebar is divided into 'WORKSPACE' and 'ADMINISTRATION' sections. Under 'WORKSPACE', 'Funding' is selected. The main content area shows the 'Funding' page with a sub-header 'Funding sources and arrangements.' and a description: 'Funding arrangements are held per student and tracked over time (a change closes the current arrangement and opens a new one). Manage them on a student's page. Below are the funding sources available.' A table titled 'Funding sources' lists two sources: 'UKRI EPSRC research council' and 'University Scholarship Fund university'. A bottom right button says 'Ask PGR'.

Funding arrangements. Approved stipends emit events to Finance without leaving this table.

Funding integrity

What it solves. The cohort-wide health check for funding. Runs eight integrity checks per student (gap between arrangements, over-payment, missing project link, expiring within threshold, etc.) and lists every student whose chain has a problem, **errors first**. Fast: 266 students in 0.17 s, from a bulk load — the naïve per-student version took 14× as long.

How to use it.

- Open **Funding integrity** from the Administration sidebar.
- Filter by error level; click a row to jump into the student's funding card.
- Zero errors is the target after any bulk import of funding arrangements.

Errors only

 Ask PGR

Page 16

Thesis

What it solves. The examination pipeline: **intention** → **submit** → **nominate examiners** → **viva** → **outcome** → **corrections**. Examiners carry an *affiliation*; the **conflict-of-interest** check blocks approving an examiner who has co-authored with the candidate or the supervisor. Corrections open at outcome time with the correct clock — minor 28 days, major 182 days — and require sign-off.

How to use it.

- Filter theses by stage (submitted, under exam, awaiting corrections, complete).
- Nominate examiners from the thesis detail; the platform will refuse a nominee that fails the Col test.
- After the viva, record the outcome. Corrections tasks appear automatically on the student's queue.

The screenshot shows the PGR Platform interface. The top navigation bar includes the PGR Platform logo, a search bar, and a 'Ctrl K' button. The left sidebar is divided into 'WORKSPACE' and 'ADMINISTRATION' sections. The 'WORKSPACE' section includes links to Dashboard, Analytics, My journey, Tasks, Persons, Research, Recruitment, Admissions, Students, Supervision, Progression, Funding, Thesis (highlighted), and Completion. The 'ADMINISTRATION' section includes links to Funding integrity, Statutory, and Workflows. The main content area is titled 'Thesis & examination' and contains a 'Students' table. The table has two columns: 'Student' and 'Status'. The 'Student' column lists various students with IDs like 'SCALE-99-0300', 'SCALE-99-0299', 'SCALE-99-0298', 'SCALE-99-0297', 'SCALE-99-0296', 'SCALE-99-0295', 'SCALE-99-0294', 'SCALE-99-0293', 'SCALE-99-0292', 'SCALE-99-0291', and 'SCALE-99-0290'. The 'Status' column shows 'active' or 'withdrawn' for each student. An 'Open' button is visible next to each row. A blue 'Ask PGR' button is located at the bottom right of the table.

Student	Status
SCALE-99-0300	active
SCALE-99-0299	withdrawn
SCALE-99-0298	withdrawn
SCALE-99-0297	withdrawn
SCALE-99-0296	active
SCALE-99-0295	active
SCALE-99-0294	active
SCALE-99-0293	active
SCALE-99-0292	withdrawn
SCALE-99-0291	registered
SCALE-99-0290	active

Theses list. The Col block is a hard rule, not a warning.

Completion

What it solves. Graduation is one atomic transaction: **award recorded, funding closed, student identity closed, alumnus identity opened**. All under the same `person_id`. This is where you convene ceremonies and confirm final awards.

How to use it.

- Filter by ceremony or by cohort.
- Click **Graduate** on a completion row to fire the whole atomic transaction.
- Alumnus records appear in Persons the next time you look.

WORKSPACE

-  Dashboard
-  Analytics
-  My journey
-  Tasks
-  Persons
-  Research
-  Recruitment
-  Admissions
-  Students
-  Supervision
-  Progression
-  Funding
-  Thesis

 Completion

ADMINISTRATION

-  Funding integrity
-  Statutory
-  Workflows

 admin
admin@example.com > Completion

Completion & graduation

Confirm completion and graduate.

 Students

Confirm completion and graduate on the student's page. Graduation records the award, closes funding, marks the student completed, and opens the alumni identity.

Student	Status	
SCALE - 99 - 0300	active	Open
SCALE - 99 - 0299	withdrawn	Open
SCALE - 99 - 0298	withdrawn	Open
SCALE - 99 - 0297	withdrawn	Open
SCALE - 99 - 0296	active	Open
SCALE - 99 - 0295	active	Open
SCALE - 99 - 0294	active	Open
SCALE - 99 - 0293	active	Open
SCALE - 99 - 0292	withdrawn	Open
SCALE - 99 - 0291	registered	Open
SCALE - 99 - 0290	active	Open

 Ask PGR

Completions queue. The Graduate action is irreversible without database surgery — check first.

4 · Insight and administration

Analytics — Enterprise 360

What it solves. Five lenses on the whole PGR estate: **Volumes, Progression, Funding, Completion, Risk**. Risk scoring is **explainable** — every at-risk student comes with the rule-based reasons that produced the score, so a human can validate or override before acting.

How to use it.

- Pick the lens from the tabs at the top.
- Hover the risk tiles for the rules that fired.
- Click a student in the at-risk list to jump into their record.

The screenshot shows the PGR Platform Analytics interface. The sidebar on the left contains navigation options under 'WORKSPACE' and 'ADMINISTRATION'. The main content area is titled 'Analytics' and 'PGR Enterprise 360, risk, and forecasting.' It features a 'Risk & completion' section with five summary cards: 'STUDENTS AT RISK' (264, 99.2% of active), 'ON TRACK' (2, active, no risk flag), 'COMPLETIONS' (31, 9% rate), 'AVG TIME TO COMPLETION' (—, graduated students), and 'ACTIVE RESEARCHERS' (266, registered / active). Below these cards is a list of 'AT-RISK STUDENTS' with their names, UAT IDs, and funding status (e.g., 'no active funding', 'milestone overdue'). A search bar and a 'Ctrl K' button are visible at the top right of the main content area.

Enterprise 360. Read-replica routing keeps analytical reads off the write path.

Statutory returns

What it solves. Statutory exports (HESA and similar) as **versioned configuration**, not code. A **report profile** is a set of field mappings for a specific academic year; you can clone a profile to a new year, edit the mappings, validate against real students, and generate the CSV. The ~12 fields shipped are a worked example — a specialist must confirm names and coding for a real return.

How to use it.

- Pick a profile (or clone last year's).
- Edit any field mapping using dotted paths against the student model.
- Click **Validate** to run every student against every field; fix any errors listed.

- Click **Generate** — the CSV is queued as an export job under Integration.

PGR Platform
RESEARCH LIFECYCLE

WORKSPACE

Dashboard

Analytics

My journey

Tasks

Persons

Research

Recruitment

Admissions

Students

Supervision

Progression

Funding

Thesis

Completion

ADMINISTRATION

Funding integrity

Statutory

Workflows

admin
admin@example.com

FUSION
PRACTICES

ORACLE
Partner

PGR Platform

Q Search or jump to...

Ctrl K

Statutory

Statutory returns

+ New profile

A statutory return is configuration, not code — HESA is an external specification, expressed as a versioned profile of field mappings.

Report profiles

Versioned by academic year. Next year's change is an edit here, not a code release.

Code	Name	Academic year	Version	Fields	Status
HESA_STUDENT	HESA Student Return	2026/27	v1	12	active

HESA_STUDENT — 2026/27

+ Add field

Validate

Generate

Clone to new year

Target field ← source expression + transform + validation.

#	Target field	Source expression	Transform	Required	Allowed values	Default
1	HUSID	student.ref	—	required	—	—
2	SURNAME	person.familyName	upper	required	—	—
3	FNAMES	person.givenName	upper	required	—	—
4	NATION	person.nationality	upper	—	—	—
5	COURSEID	programme.code	—	required	—	—
6	INSTUFE	funding.type	lower	—	—	—
7	STULOAD	student.mode	lower	required	full_time, part_time	—
8	COMDATE	student.startDate	date_compact	required	—	—
9	ENDDATE	student.expectedEndDate	date_compact	—	—	—

Ask PGR

Statutory profiles. Every mapping is a dotted path, never an expression — safer to audit.

Workflows

What it solves. The workflow engine that ships around the state machine — task creation, notification, and outbox emission all happen in the same transaction as the state change. This screen shows the configured definitions (which events raise which tasks and notifications) and each instance's timeline.

How to use it.

- Browse workflow definitions on the left.
- Click an instance on the right for its task, notification and outbox trail.

Workflow definitions and instances. All work is transactional — no half-fired tasks.

Tasks

What it solves. Your personal work queue. Every workflow (a stipend due, a milestone overdue, a viva to schedule) raises a task that lands here for the right person or role.

How to use it.

- Filter by status (open, due today, overdue).
- Click a task to jump straight into the record it is about.
- Mark done from either the task list or the underlying record.

PGR Platform

RESEARCH LIFECYCLE

FUSION PRACTICES

ORACLE Partner

PGR Platform

Q Search or jump to...

Ctrl K

WORKSPACE

Dashboard

Analytics

My journey

Tasks

Persons

Research

Recruitment

Admissions

Students

Supervision

Progression

Funding

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ADMINISTRATION

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→

Tasks

Tasks & notifications

Work assigned to you and your roles.

My task queue

Plan funding renewal — arrangement expires 2026-10-19	PGR Administrator	Complete
Plan funding renewal — arrangement expires 2026-10-29	PGR Administrator	Complete
Plan funding renewal — arrangement expires 2026-10-22	PGR Administrator	Complete
Plan funding renewal — arrangement expires 2026-09-28	PGR Administrator	Complete
Plan funding renewal — arrangement expires 2026-08-25	PGR Administrator	Complete
Plan funding renewal — arrangement expires 2026-09-14	PGR Administrator	Complete
Plan funding renewal — arrangement expires 2026-10-14	PGR Administrator	Complete
Plan funding renewal — arrangement expires 2026-09-29	PGR Administrator	Complete
Plan funding renewal — arrangement expires 2026-09-02	PGR Administrator	Complete
Plan funding renewal — arrangement expires 2026-09-12	PGR Administrator	Complete
Plan funding renewal — arrangement expires 2026-09-18	PGR Administrator	Complete
Plan funding renewal — arrangement expires 2026-10-27	PGR Administrator	Complete
Plan funding renewal — arrangement expires 2026-10-29	PGR Administrator	Complete

Ask PGR

ASK

Tasks. Overdue tasks are escalated by the worker on its default 60-second cycle.

PGR Platform — User Manual

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Integration hub

What it solves. The nervous system for outbound events (funding to Finance, updates to Research Office) and inbound signed webhooks. See **outbox dispatcher** status, **scheduled jobs**, **exports**, the **integration log**, and the **dead-letter queue** for events that exhausted their retries.

How to use it.

- Watch the dispatcher lag — anything over a minute is worth investigating in the worker log.
- Replay a dead-letter row after fixing the partner endpoint.
- Signed webhooks are validated by HMAC-SHA256 against the raw body; malformed payloads land as *failed*, never dropped.

The screenshot shows the PGR Platform interface. The top navigation bar includes the PGR Platform logo, a search bar, and a notification bell. The left sidebar lists various workspace and administration options. The main content area is titled 'Integration hub' and displays the status of the integration boundary as 'healthy'. It shows three categories of events: PENDING (0), DISPATCHED (IN WINDOW) (5), and DEAD-LETTERED (0). Below this, there is a table for 'TRAFFIC BY SYSTEM' and a section for 'FAILED INBOUND MESSAGES'.

System	Direction	Status	Count
finance	inbound	success	1
finance	outbound	success	2
hr	inbound	success	1
internal	outbound	skipped	3
research	inbound	success	1

Integration hub. This is where you find out that Finance hasn't been receiving anything.

Audit

What it solves. Immutable record of every privileged action. Actor is resolved from the JWT with no DB hit, so this does not slow the request path. Filter by user, action, entity or request ID.

How to use it.

- Filter by any of the four columns.
- Copy a request ID and search server logs to find the exact line for that action.
- Trails are never deleted — even after a person leaves.

WORKSPACE

- Dashboard
- Analytics
- My journey
- Tasks
- Persons
- Research
- Recruitment
- Admissions
- Students
- Supervision
- Progression
- Funding
- Thesis
- Completion

ADMINISTRATION

- Funding integrity
- Statutory
- Workflows

admin
admin@example.com



🏠 > Audit

Audit

Immutable record of privileged actions and state changes.



Filters

Entity type

Entity ID

Actor email

Time	Actor	Method	Action	Entity	Status
2026-08-22 11:58:14	admin@example.com	POST	POST /api/v1/pattern-lab/models/c882eadc-c184-4317-8eda-4ca3d54dc2b3/retrain	pattern-lab - c882eadc	200
2026-08-22 11:55:06	admin@example.com	POST	POST /api/v1/pattern-lab/models/c882eadc-c184-4317-8eda-4ca3d54dc2b3/retrain	pattern-lab - c882eadc	200
2026-08-22 11:14:18	admin@example.com	POST	POST /api/v1/pattern-lab/versions/9b18c948-d2d2-42b9-81f0-24dbc80e28fc/transition	pattern-lab - 9b18c948	200
2026-08-22 11:13:40	admin@example.com	POST	POST /api/v1/pattern-lab/versions/9b18c948-d2d2-42b9-81f0-24dbc80e28fc/transition	pattern-lab - 9b18c948	200
2026-08-22 11:13:06	admin@example.com	POST	POST /api/v1/pattern-lab/versions/9b18c948-d2d2-42b9-81f0-24dbc80e28fc/transition	pattern-lab - 9b18c948	200
2026-08-22 10:44:35	admin@example.com	POST	POST /api/v1/pattern-lab/datasets/a1b254b2-4afc-4765-83bd-d5abdfc35aa7/discover	pattern-lab - a1b254b2	200
2026-08-22 10:44:27	admin@example.com	POST	POST /api/v1/pattern-lab/datasets	pattern-lab	201
2026-08-22 10:23:39	admin@example.com	POST	POST /api/v1/pattern-lab/datasets/fcbe4914-ecb9-4748-926d-f28df09f118a/discover	pattern-lab - fcbe4914	200
2026-08-22 10:23:26	admin@example.com	POST	POST /api/v1/pattern-lab/datasets	pattern-lab	200
2026-08-22 09:03:13	admin@example.com	POST	POST /api/v1/research/supervisor-suggestions	research	200

Ask PGR

25K

Audit trail. Every mutation is here; retention is forever.

Settings — real institutional configuration

What it solves. Not just personal preferences. Change a value here and the platform behaves differently on the very next request. Wired settings include the **supervision capacity limit**, the **expected meeting interval** (drives overdue flags), the **part-time factor** (drives mode-change recalculation), the **minimum funding gap**, the **email kill-switch**, the **assistant LLM enable flag**, and CRUD for departments, areas, programmes and funding sources.

How to use it.

- **Institution** tab: change a value; it validates by type and range, and the reset button removes the override.
- **List of values** tab: add/rename/deactivate departments, areas, programmes, sources. Delete is refused while in use, telling you exactly what still references the value.
- **Users & roles** tab: invite a user (the invitation *is* the password-reset email), activate/deactivate, assign roles. You cannot lock yourself out.

The screenshot displays the PGR Platform Settings interface. The left sidebar contains a 'WORKSPACE' section with links to Dashboard, Analytics, My journey, Tasks, Persons, Research, Recruitment, Admissions, Students, Supervision, Progression, Funding, Thesis, and Completion. Below this is an 'ADMINISTRATION' section with links to Funding integrity, Statutory, and Workflows. The main content area is titled 'Settings' and includes a sub-header 'Institution configuration, reference data, user administration and your personal preferences.' It features four tabs: 'List of values' (selected), 'Institution policy', 'Users & roles', and 'My preferences'. Under the 'List of values' tab, there are four sub-tabs: 'Departments' (selected), 'Research areas', 'Programmes', and 'Funding sources'. A message states: 'Values still referenced by live records cannot be deleted — the platform will tell you exactly what is using them.' Below this is a table with the following data:

Name	Code	In use	Actions
Computer Science	CS	399 records	

At the bottom of the table, there is a link: '> Platform-fixed value sets read-only'. The bottom of the page shows a user profile for 'admin' and an 'Ask PGR' button.

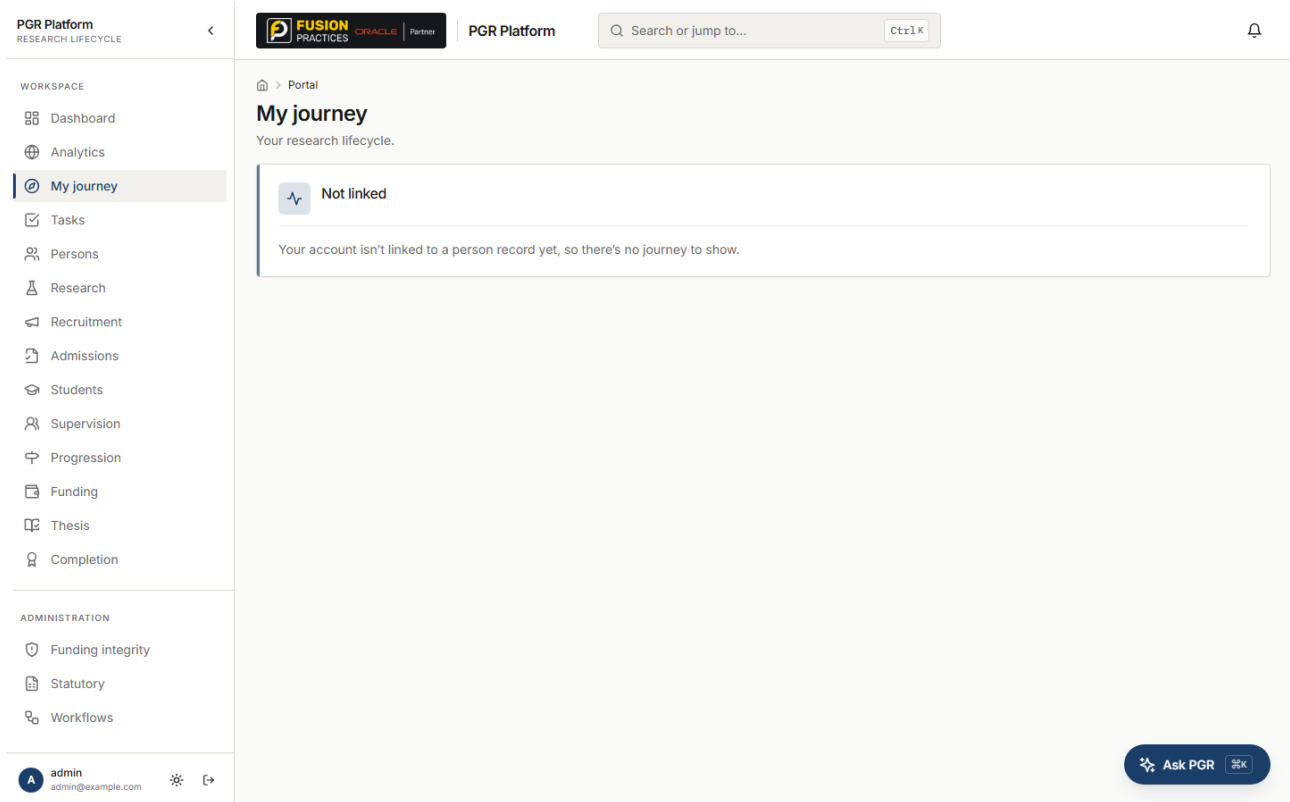
Settings. The Institution tab is the institution's control panel; the Users tab is the identity centre.

Student portal — My journey

What it solves. The student's own view of themselves. Milestones, meetings, funding schedule, thesis progress, and any tasks they need to complete. Row-scoped so a student can only see their own record; the supervisor version at **/portal** presents the caseload similarly.

How to use it.

- Sign in as the student to see this screen.
- The action buttons here (log meeting, upload document, request extension) are the same ones an administrator uses from the student record.



The student's self-service portal — every action is audited.

Pattern Lab (Advanced)

What it solves. Governed institutional ML. Full pipeline: **Discover** → **Validate** → **Train** → **Approve** → **Deploy** → **Monitor** → **Retrain**. Approver separation is enforced; nothing auto-promotes to production. Every prediction carries the *perturbation factors* (how much each feature moved the score) and the score sits *beside* the deterministic risk indicators, never replacing them.

How to use it.

- **Targets** tab: pick a governed target. Insufficient data is refused with the missing tables named.
- **Training**: bounded candidate search, stratified 5-fold CV. A run that cannot beat baseline is reported *failed*.
- **Governance**: an approver (separate from the trainer) promotes candidate → production.
- **Monitoring**: PSI drift, AUC vs actuals, health verdict — the system recommends review, never acts.

The screenshot displays the PGR Platform interface, specifically the Pattern Lab section. The left sidebar contains navigation links for Workspace (Dashboard, Analytics, My journey, Tasks, Persons, Research, Recruitment, Admissions, Students, Supervision, Progression, Funding, Thesis, Completion) and Administration (Funding integrity, Statutory, Workflows). The main content area is titled 'Pattern Lab' and includes a search bar and a 'New analysis' button. The 'Overview' tab is selected, showing 'Active models' (2 models) and 'Recent discoveries'. The 'Active models' section displays two models: 'Progression Delay Risk model' (candidate, Gradient boosting, AUC 0.609) and 'Funding Continuity Risk model' (candidate, Gradient boosting, AUC 0.702). The 'Recent discoveries' section shows two discoveries: 'Students with stipend amount at or below 18250 experienced a funding gap at 36%, vs 22% for above 18250 — 1.7x the rate.' (significant, 1.7x the rate, Funding). The 'Data health' section shows 'Whether each governed question has enough labelled history to analyse honestly.' and 'Ask PGR' button. The bottom of the interface shows the user 'admin' and the email 'admin@example.com'.

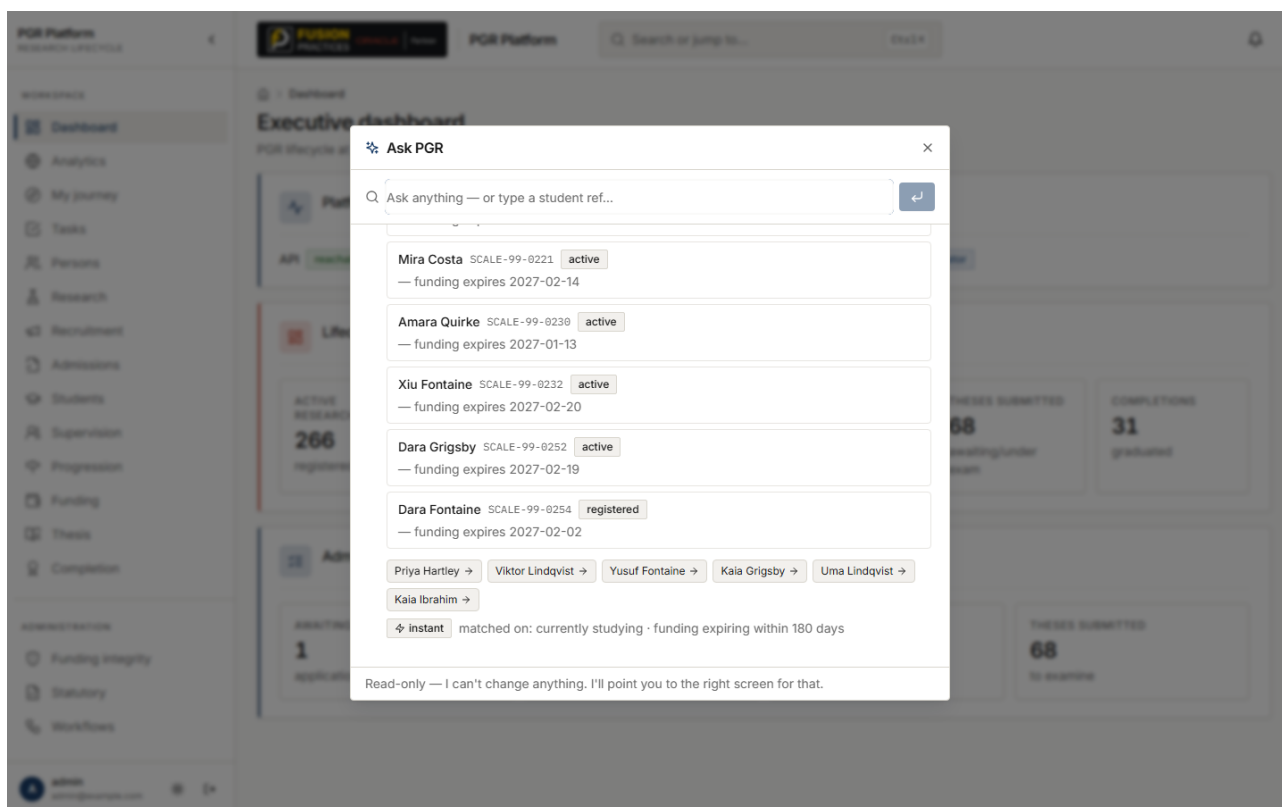
Pattern Lab. The current live model caught itself: matured AUC 0.48 vs trained 0.70 → surfaced for review.

Ask PGR — natural-language layer

What it solves. A rules-first assistant. Ask a question in your own words — *who is falling through the cracks, whose money is running out, no supervision meeting in 90 days* — and the platform understands, executes the equivalent filter, and shows the students. The LLM is off by default; the concept graph and typo-corrected parser cover most real questions with **no student data leaving the server**. Read-only in this release.

How to use it.

- Launch it from the floating **Ask PGR** button or press **Ctrl/Cmd+K**.
- Type your question; the panel shows results as you type.
- Every answer carries an **Understood as** readback so you can verify the interpretation.
- Click a suggested student to open their record; the assistant does not do writes.



Ask PGR answering "funding expiring in 6 months" — matched deterministically in ~1 ms.

Appendix · Environment and operations

Batch files in the repo root:

start-all.bat	Postgres + backend + worker + frontend + open browser
start-backend.bat	FastAPI on http://127.0.0.1:8000
start-worker.bat	Background worker (scheduler + outbox + notifications)
start-frontend.bat	Next.js on http://localhost:3000
start-postgres.bat	Start the local Postgres 18 service
stop.bat	Stop everything the batch files started
test.bat	Run backend test suite (224 tests)

Key environment variables (backend/ .env):

DATABASE_URL / DATABASE_REPLICA_URL	Primary and read-replica connection strings
STORAGE_BACKEND / STORAGE_ROOT	Where document uploads live (local FS in dev)
EMAIL_BACKEND	console smtp — the wire-format for notifications
OUTBOX_MAX_ATTEMPTS	How many times to retry a failed webhook before dead-letter
MAX_FAILED_LOGINS / LOCKOUT_MINUTES	Account lockout policy
ASSISTANT_LLM_ENABLED	Turn the optional LLM path in Ask PGR on/off
APP_SECRET_KEY	Signs access/refresh JWTs and inbound webhook HMAC

End of manual. Companion document `PGR_Platform_Critical_Gaps.pdf` lists the things that look implemented but rely on workarounds — read it before promoting the platform to a real institution.